№5 23.10.2024

$\boxed{1}$

Дискретный случай

Дискретные случаи

$$W = \int_{\Omega} F(u, p_1, p_2)\, d\Omega \qquad p_1 = \frac{\partial u}{\partial x^1},\ p_2 = \frac{\partial u}{\partial x^2}$$

В-р М

$$W_h = \sum_{\Omega_h} F(\underset{u_h}{\underset{\|}{d_0^+ u}}, \underset{p_{1h}}{\underset{\|}{d_1^+ u}}, \underset{p_{2h}}{\underset{\|}{d_2^+ u}})\, \Delta V$$

$$\delta W_h = \Delta V \sum_{\Omega} \frac{\partial F}{\partial u_h} \delta(d_0^+ u) + \frac{\partial F}{\partial p_{1h}} \delta(d_1^+ u) + \frac{\partial F}{\partial p_{2h}} \delta(d_2^+ u) = 0$$

$$\sum_{\Omega_h} \left[\frac{\partial F}{\partial u_h} d_0^+ \delta u + \frac{\partial F}{\partial p_{1h}} d_1^+ \delta u + \frac{\partial F}{\partial p_{2h}} d_2^+ \delta u\right] = 0$$

Сеточное $\int$-е по частям

$\int$ по частям

по опред. сопр. опер.

$$\sum_{(i,j)\in\Omega_h} f_{ij} (d_m^+ g)_{ij} = \sum_{(i,j)\in\Omega_h} \left[(d_m^+)^* f\right]_{ij} g_{ij}$$

$$(d_0^+)^* = d_0^-,\ (d_1^+)^* = -d_1^-,\ (d_2^+)^* = -d_2^-$$

$$\sum_{\Omega_h} \left(d_0^- \frac{\partial F}{\partial u_h}\right)\delta u - \left(d_1^- \frac{\partial F}{\partial p_{1h}}\right)\delta u - \left(d_2^- \frac{\partial F}{\partial p_{2h}}\right)\delta u = 0 \Rightarrow$$

$$\sum_{\Omega_h} \left(d_0^- \frac{\partial F}{\partial u_h} - d_1^- \frac{\partial F}{\partial p_{1h}} - d_2^- \frac{\partial F}{\partial p_{2h}}\right)\delta u = 0 \Rightarrow \left| d_0^- \frac{\partial F}{\partial u_h} - d_1^- \frac{\partial F}{\partial p_{1h}} - d_2^- \frac{\partial F}{\partial p_{2h}} = 0\right.$$

аналог ур-е эйлера

$\boxed{2}$

Ур-е Эйлера	Разностная схема
$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x^1}\frac{\partial F}{\partial p_1} - \frac{\partial}{\partial x^2}\frac{\partial F}{\partial p_2} = 0$	$d_0^- \frac{\partial F}{\partial u_h} - d_1^- \frac{\partial F}{\partial p_{1h}} - d_2^- \frac{\partial F}{\partial p_{2h}} = 0$

φ

Пример

Ур-е ~~Лапласа~~ Пуассона

$+ u\varphi(x^1, x^2)\,dx$

$$W = \frac{1}{2}\int_\Omega \left(\frac{\partial u}{\partial x^1}\right)^2 + \left(\frac{\partial u}{\partial x^2}\right)^2 + u\cdot\varphi(x^1 x^2)\,dx$$

$u\varphi$ and x

ΔS: площадь ячеек

$$W_h = \frac{1}{2}\sum_{\Omega_h}\left[(d_1^+u)^2 + (d_2^+u)^2 + d_0^+u\, d_0^+\varphi\right]\Delta S$$

$$\delta W_h = \frac{\Delta S}{2}\sum_{\Omega_h} 2(d_1^+u)(d_1^+\delta u) + 2(d_2^+u)(d_2^+\delta u) + (d_0^+\delta u)(d_0^+\varphi) = 0$$

↓ сеточное ∫-е по частям

$$\delta W_h = \sum_{\Omega_h} -(d_1^- d_1^+ u)\,\delta u - (d_2^- d_2^+ u)\,\delta u + \frac{1}{2}(d_0^- d_0^+\varphi)\,\delta u = 0$$

$$\Rightarrow \sum_{\Omega_h}\left[-d_1^- d_1^+ u - d_2^- d_2^+ u + \frac{1}{2} d_0^- d_0^+ \varphi\right]\delta u = 0$$

d^-d^+ сопрозиные оператор

$\Rightarrow$ разностная схема $\boxed{d_1^- d_1^+ u + d_2^- d_2^+ u = \frac{1}{2} d_0^- d_0^+ \varphi}$ ← сеточное ур-е Пуассона ← разностная

$$\Rightarrow (d_1^+ f)_{ij} = \frac{1}{2h_1}\left(u_{i+1 j} + u_{i+1 j+1} - u_{ij} - u_{ij+1}\right)$$

$$(d_1^- f)_{ij} = \frac{1}{2h_1}\left(-u_{i-1 j} - u_{i-1 j-1} + u_{ij} + u_{i j-1}\right)$$

d_2 

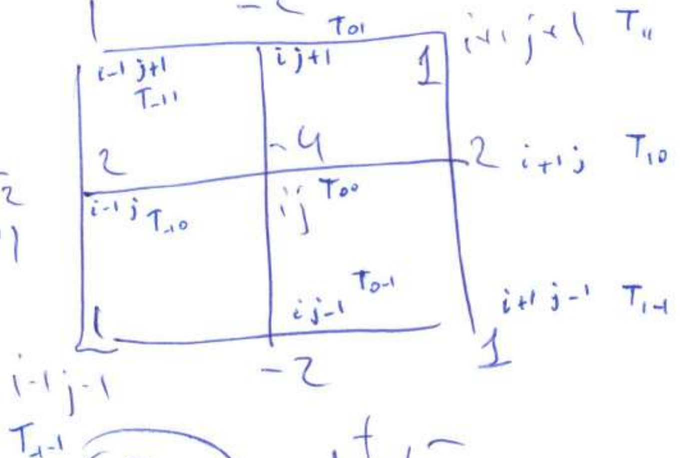
$$d_1^+ = \frac{1}{2h_1}\left(T_{10} + T_{11} - T_{00} - T_{01}\right)$$

16 слаг.

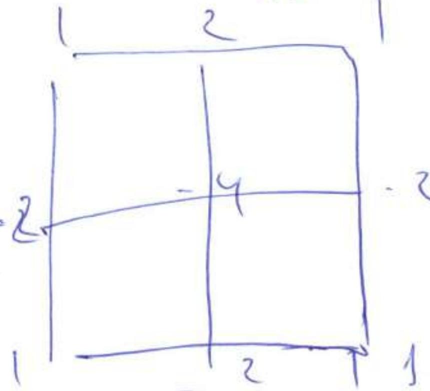
$$d_1^- = \frac{1}{2h_1}\left(-T_{-10} - T_{-1-1} + T_{00} + T_{0-1}\right)$$

$$d_1^- d_1^+ = \frac{1}{4h_1^2}\Big[T_{10}\left(-T_{-10}\right) + T_{10}\left(-T_{-1-1}\right) + \ldots + -T_{01}T_{0-1}\Big]$$

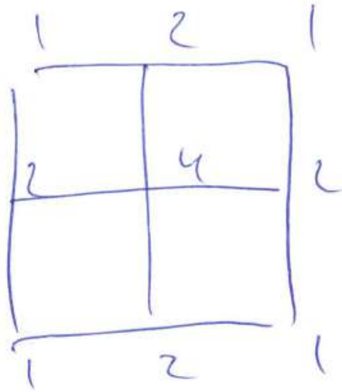
$$= \frac{1}{4h_1^2}\Big[\underset{\substack{\| \\ T_{00}}}{-T_{1-1,\,0+0}} + \underset{\substack{\| \\ T_{0-1}}}{T_{1-1,\,0-1}} + \ldots - \underset{\substack{\| \\ T_{00}}}{T_{0+0,\,1-1}}\Big]$$

$\frac{1}{4h_1^2}$ 

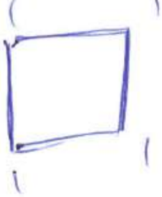
$D_{11} = d_1^+ d_1^-$

$\frac{1}{4h_2^2}$ 

$D_{22} = d_2^+ d_2^-$
$= d_2^- d_2^+$

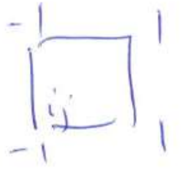
$D_{00} = \frac{1}{16}$ 

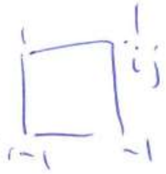
$d_0^- d_0^+$

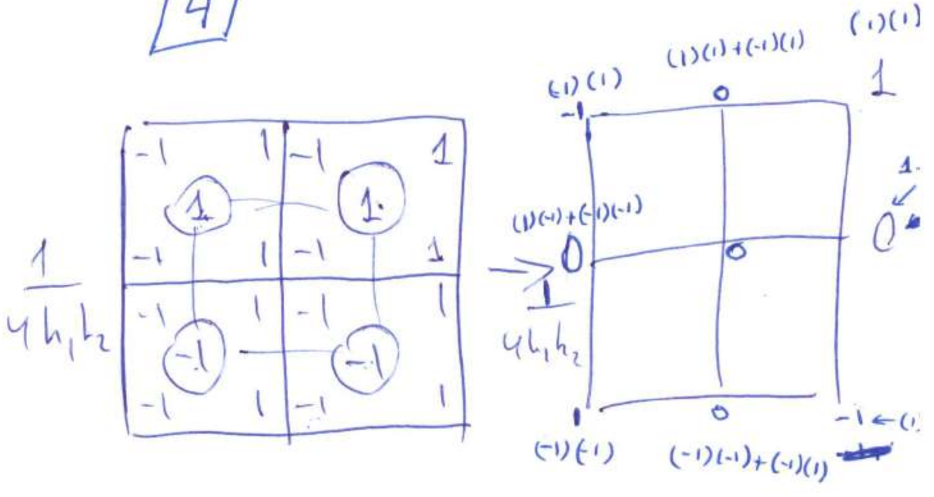
$d_0^+ = \frac{1}{4}$ 

$\boxed{4}$

$D_{12} = d_1^+ d_2^-$

$d_1^+ = \frac{1}{2h_1}$ 

$d_2^- = \frac{1}{2h_2}$ 

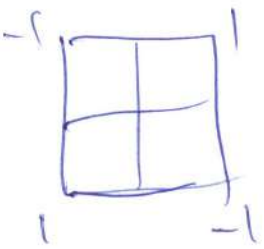


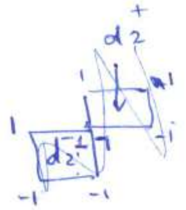
$$D_{12} = \frac{1}{4h_1h_2}\left(u_{i+1\,j+1} + u_{i-1\,j-1} - u_{i+1\,j-1} - u_{i-1\,j+1}\right)$$

в центре

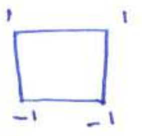
$(1)(-1)+(1)(1)+(-1)(-1)+(1)(-$

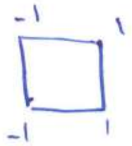
$-1+1+1-1=0$

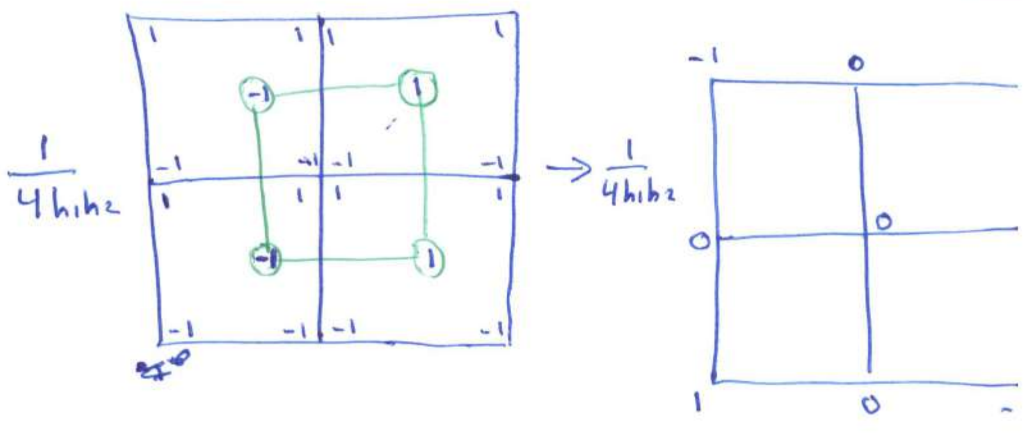




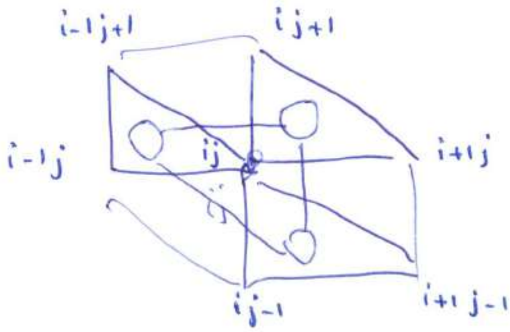
$D_{21} = d_2^+ d_1^-$

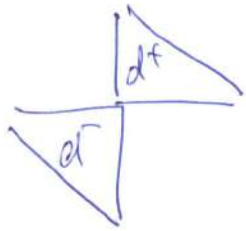
$d_2^+ = \frac{1}{2h_2}$ 

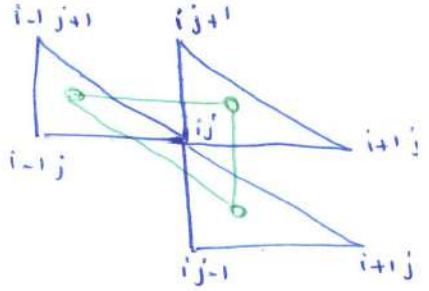
$d_1^- = \frac{1}{2h_1}$ 

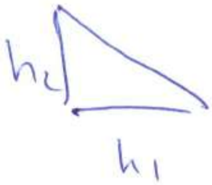


$$D_{21} = \frac{1}{4h_1h_2}\left(u_{i+1\,j+1} + u_{i-1\,j-1} - u_{i+1\,j-1} - u_{i-1\,j+1}\right)$$



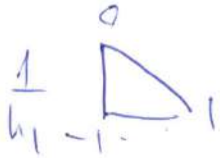






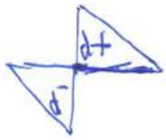
$$(d_0^+ f)_{ij} = \frac{1}{3}(f_{ij} + f_{i+1j} + f_{ij+1})$$

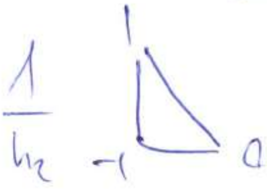
$$(d_0^- f)_{ij} = \frac{1}{3}(f_{ij} + f_{i-1j} + f_{ij-1})$$



$$(d_1^+ f)_{ij} = \frac{1}{h_1}(f_{i+1j} - f_{ij})$$

$$(d_1^- f)_{ij} = \frac{1}{h_1}(f_{ij} - f_{i-1j})$$



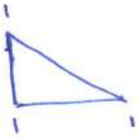


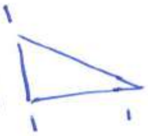
$$(d_2^+ f)_{ij} = \frac{1}{h_2}(f_{ij+1} - f_{ij})$$

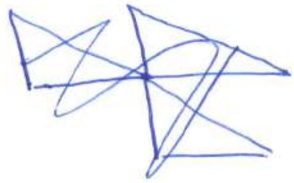
$$(d_2^- f)_{ij} = \frac{1}{h_2}(f_{ij} - f_{ij-1})$$

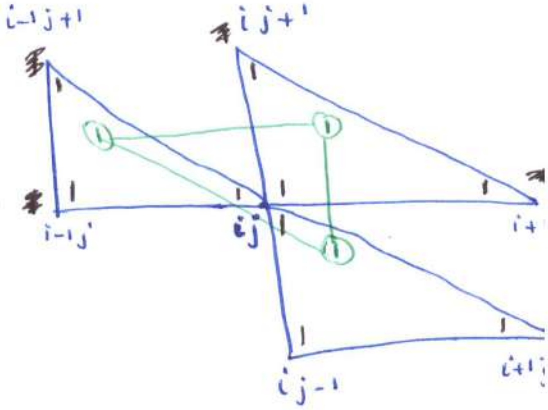
$D_{00}, D_{11}, D_{22}, D_{12}, D_{21}$

$D_{00} = d_0^+ d_0^-$

$d_0^+ = \frac{1}{3}$ 

$d_0^- = \frac{1}{3}$ 

$\Rightarrow$ 

$\frac{1}{9}$ 

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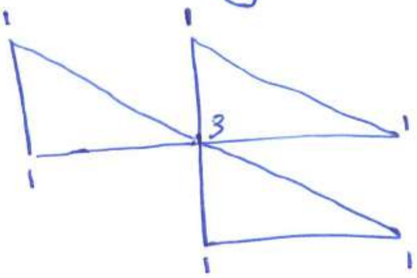
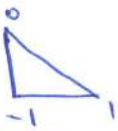
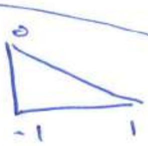


$$D_{00} = \frac{1}{9}\left(3 f_{ij} + f_{i+1\,j} + f_{i-1\,j} + f_{i\,j-1}\, f_{i+1\,j-1} + f_{i-1\,j+1} + f_{i\,j+1}\right)$$

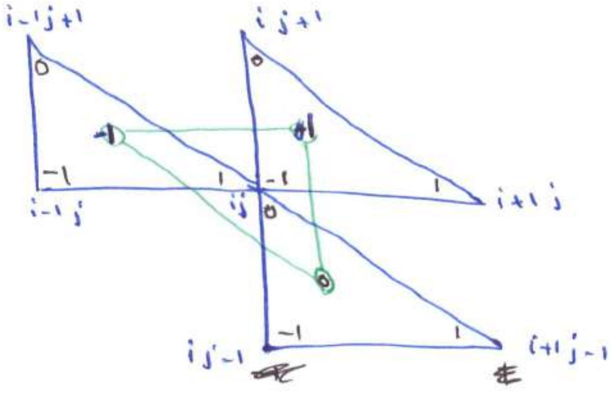
$D_{11} = d_1^+ d_1^-$

$d_1^+ = \frac{1}{h_1}$ 

$d_1^- = \frac{1}{h_1}$ 

$\Downarrow$

меняем знак

$\Rightarrow$ $\frac{1}{h_1^2}$ 

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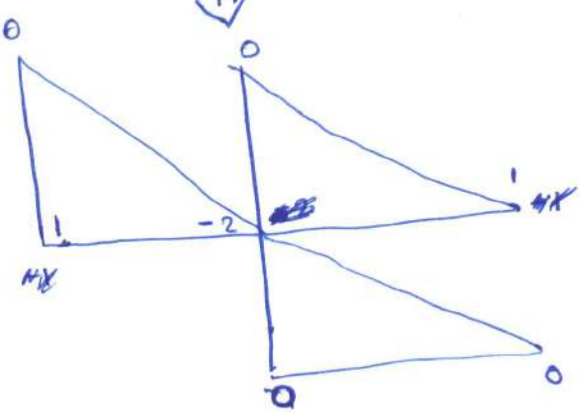
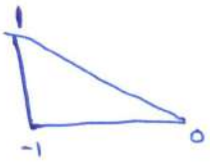
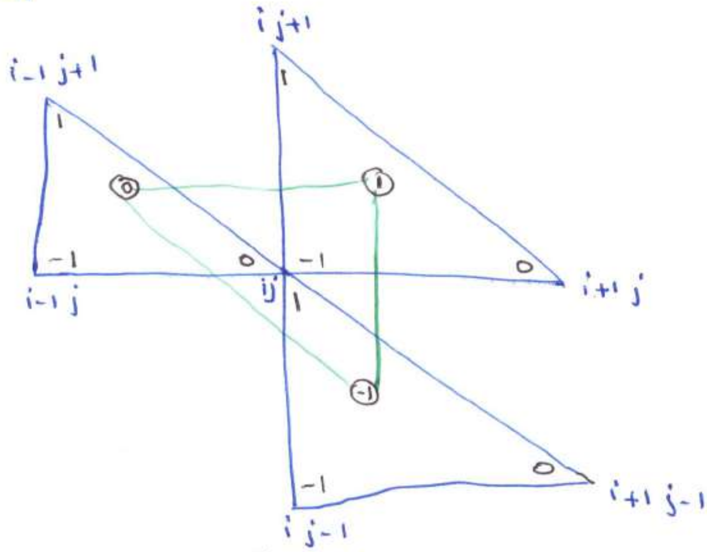


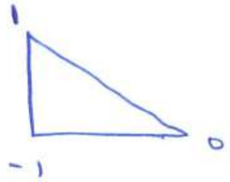
$\cancel{D_{11} = \frac{1}{h_1^2}\left(f_{i-1\,j} - f_{i+1\,j}\right.}$

$$D_{11} = \frac{1}{h_1^2}\left(f_{i-1\,j} - 2 f_{ij} + f_{i+1\,j}\right)$$

D_{22}

$d_2^+ = \frac{1}{h_2}$ 

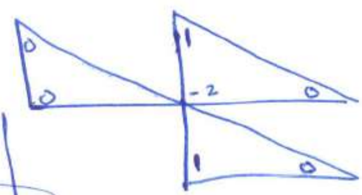
$\Rightarrow \frac{1}{h_2^2}$ 

$d_2^- = \frac{1}{h_2}$ 

$\downarrow$

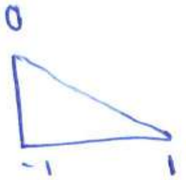
عند التدوير لـ ١٨٠ درجه
نعكس الاشارات في d^-
فقط

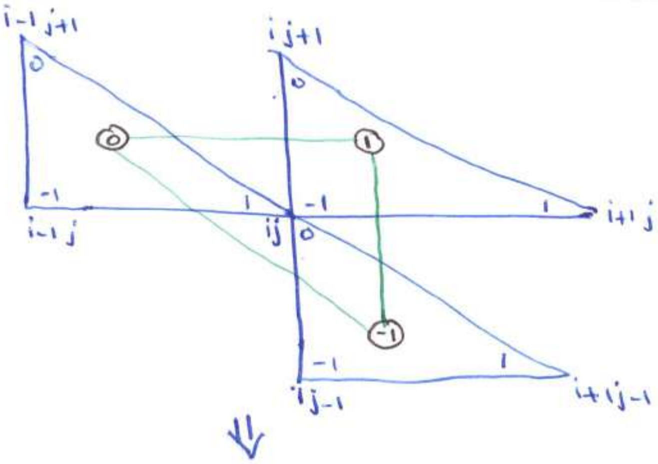
$\Downarrow$

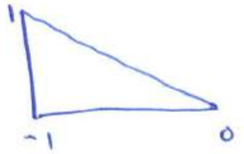


$$D_{22} = \frac{1}{h_2^2}\left(f_{i\,j+1} - 2f_{ij} + f_{i\,j-1}\right)$$

D_{12}

$d_1^+ = \frac{1}{h_1}$ 

$\Rightarrow \frac{1}{h_1 h_2}$ 

$d_2^- = \frac{1}{h_2}$ 

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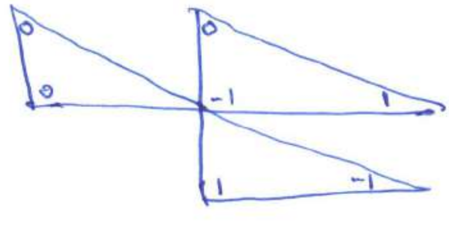
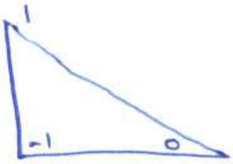


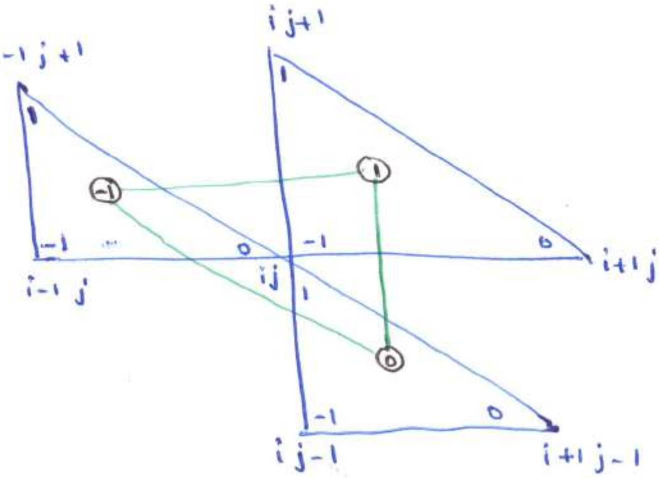
$$D_{12} = \frac{1}{h_1 h_2}\left(-f_{ij} + f_{i+1\,j} + f_{i\,j-1} - f_{i+1\,j-1}\right)$$

D_{21}

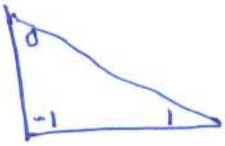
$d_2^+ = \frac{1}{h_2}$



$\Rightarrow \frac{1}{h_1 h_2}$



$d_1^- = \frac{1}{h_1}$



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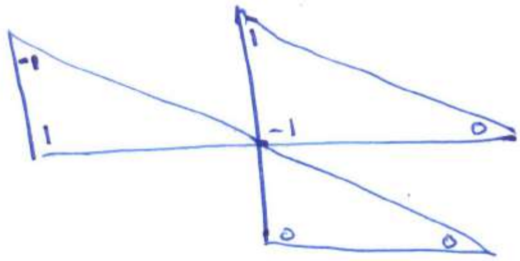

$$D_{21} = \frac{1}{h_1 h_2}\left(-f_{i-1\,j+1} + f_{i\,j+1} + f_{i-1\,j} - f_{ij}\right)$$